ENGINEERING DECISION REPORT

Logo / Branding Area

Bifacial PV Mounting System Trade-Off Report

This report compares 2-purlin, 3-purlin and 4-purlin rear-support alternatives for bifacial PV.

PROJECT Konya C&I Bifacial PV Plant	REPORT DATE 2026-03-12
REFERENCE SCENARIO 2-Purlin	RECOMMENDED SCENARIO 3-Purlin
SCENARIO COUNT 3	SITE / COORDINATES Konya, Turkiye (37.8713, 32.4846)

Report Metadata / Project Information

Context

Project Metadata

PREPARED BY **Bifacial Analytics Platform**	CLIENT **Sample Energy Investments**
PROJECT CODE **KON-CI-2026-04**	REPORT VERSION **2.0**
CURRENCY **USD**	ANNUAL ALBEDO **0.214**
ANNUAL GHI (KWH/M2) **1,848**	REFERENCE SCENARIO **2-Purlin**

Site Notes

Semi-arid climate with strong summer irradiance and moderate winter diffuse component.

Decision Context

This report is designed to support purlin-count selection by balancing energy yield, structural confidence, and indicative commercial implications.

BEST ENERGY SCENARIO

2-Purlin

Annual total: 1,096,925 kWh

BEST STRUCTURAL CONFIDENCE

3-Purlin

Score: 84.0/100

LOWEST STRUCTURE COST

2-Purlin

Cost: +452,000 USD

Executive Summary

Investor / Management

This report compares 2-purlin, 3-purlin and 4-purlin rear-support alternatives for bifacial PV. Results show that lower purlin count improves rear-side gain and energy output, while higher purlin count increases structural confidence and CAPEX. The recommended option balances annual yield advantage with acceptable structure risk and commercial efficiency.

Decision Snapshot

One-Page Decision

RECOMMENDED SCENARIO

3-Purlin

Decision score: 84.6/100

ENERGY DELTA VS REFERENCE

-3,708 kWh

Positive values indicate higher annual total energy.

CAPEX DELTA VS REFERENCE

+17,500 USD

Indicative structure CAPEX difference.

Why This Option

- Higher static margin relative to 2-purlin reference.
- Incremental CAPEX uplift requires yield-offset validation.

Watch-Outs NEGATIVE

- Recommended option carries a structure CAPEX uplift versus reference.

The recommended scenario is 2-purlin + brace due to balanced energy and structural profile.

Methodology Overview

Technical Basis

- Monte Carlo rear-side obstruction simulation was used to estimate bifacial shading impact.
- NASA POWER and PVGIS climate/albedo context was integrated.
- Scenario scoring combines energy, structure confidence and cost normalization.

Data Sources and Assumptions

Input Integrity

SOURCE	PROVIDER	RETRIEVED	NOTE
NASA POWER	NASA	2026-03-12T10:22:00Z	Surface albedo context
PVGIS TMY 5.2	European Commission JRC	2026-03-12T10:22:10Z	Irradiance and weather baseline

ASSUMPTION	VALUE	CONFIDENCE
Modeling simplification	Support members are approximated with box geometry for rear-side obstruction analysis.	**HIGH**
Cost baseline	Base structure CAPEX and purlin uplift assumptions are indicative and project-specific.	**MEDIUM**

Scenario Definitions

Alternatives

3-Purlin

- Purlin count: 3
- Cross-brace: No
- Rear blockage: 9.93%
- Structure cost: +469,500 USD

2-Purlin

- Purlin count: 2
- Cross-brace: No
- Rear blockage: 6.50%
- Structure cost: +452,000 USD

2-Purlin + Cross Brace

- Purlin count: 2
- Cross-brace: Yes
- Rear blockage: 19.84%
- Structure cost: +464,500 USD

Scenario Comparison Summary

Side-by-Side

#	SCENARIO	PURLIN	CROSS BRACE	ANNUAL TOTAL (KWH)	EFF. REAR GAIN (%)	STRUCTURAL SCORE	STRUCTURE COST	DELTA VS REF (KWH)
1	**3-Purlin**	3	No	1,093,217	9.32	84.0	+469,500 USD	-3,708
2	**2-Purlin (REF)**	2	No	1,096,925	9.69	74.5	+452,000 USD	0
3	**2-Purlin + Cross Brace**	2	Yes	1,082,622	8.26	82.0	+464,500 USD	-14,303

Bifacial Rear-Side Impact Analysis

Optical / Mismatch

Rear-Side Obstruction by Scenario

3-Purlin	9.93%
2-Purlin	6.50%
2-Purlin + Cross Brace	19.84%

Higher values indicate stronger backside shading/obstruction effect.

Effective Rear Gain (%)

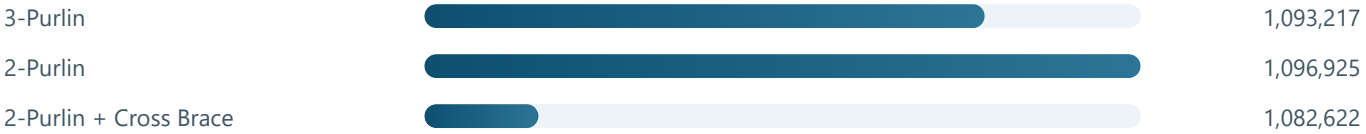
3-Purlin	9.32%
2-Purlin	9.69%
2-Purlin + Cross Brace	8.26%

Estimated effective rear-side energy contribution ratio.

Energy Yield Comparison

Yield Delta

Annual Total Energy (kWh)



Comparative annual total energy output across purlin alternatives.

Structural / Engineering Considerations

Engineering

Structural Confidence Proxy (0-100)

3-Purlin	84.0
2-Purlin	74.5
2-Purlin + Cross Brace	82.0

Proxy score intended for comparative decision support, not final static approval.

Engineering Interpretation

- Structural confidence scores are comparative indicators based on purlin count and brace presence.
- Final structural acceptance requires project code checks, geotechnical envelope, and detailed design verification.
- Reference scenario for cost deltas: 2-Purlin.

Cost and Commercial Comparison

Commercial

Indicative Structure Cost (USD)

3-Purlin	469,500
2-Purlin	452,000
2-Purlin + Cross Brace	464,500

Structure CAPEX proxy driven by purlin count and brace usage assumptions.

SCENARIO	STRUCTURE COST	DELTA VS REF	LCOE DELTA (%)
3-Purlin	+469,500 USD	+17,500 USD	N/A
2-Purlin	+452,000 USD	0 USD	N/A
2-Purlin + Cross Brace	+464,500 USD	+12,500 USD	N/A

Uncertainty / Sensitivity Interpretation

Risk Envelope

Monte Carlo-based results are interpreted as confidence-aware scenario outcomes. Values indicate decision-oriented ranges, not deterministic guarantees.

SCENARIO	P90 (KWH)	P50 (KWH)	P10 (KWH)	INTERPRETATION
3-Purlin	1,089,700	1,093,217	1,096,900	Moderate spread
2-Purlin	1,093,400	1,096,925	1,100,500	Narrow uncertainty envelope
2-Purlin + Cross Brace	1,078,700	1,082,622	1,086,600	Higher spread due to stronger rear-side obstruction

Key Trade-Offs

Decision Drivers

- 2-purlin has better rear-side gain and lower CAPEX but lower structural confidence proxy.
- 3-purlin improves structural confidence with modest energy penalty and CAPEX increase.
- Cross-brace option improves structural behavior but can materially reduce rear-side gain.

Final Recommendation

Decision

RECOMMENDED SCENARIO

3-Purlin

3-purlin is recommended for this site profile due to balanced yield retention and stronger structural confidence.

- Energy loss versus 2-purlin remains limited while structure confidence increases materially.
- Cost uplift is manageable under current CAPEX assumptions and likely acceptable for EPC risk control.

If geotechnical verification confirms low structural risk, 2-purlin can be selected for maximum energy-driven return.

Investor Interpretation: 3-purlin provides a more bankable risk-return balance for financing discussions.

Appendix / Technical Notes

Reference

- Detailed static checks are outside this report scope and should be completed by structural engineering.
- Cost assumptions are indicative placeholders and must be updated with procurement quotations.
- Simulation console extract is included for traceability.

SCENARIO ID	LABEL	PURLIN	CROSS BRACE	REAR BLOCKAGE (%)	MISMATCH (%)	DECISION SCORE
s3	3-Purlin	3	No	9.93	13.75	84.6
s2	2-Purlin	2	No	6.50	13.62	83.1
s2b	2-Purlin + Cross Brace	2	Yes	19.84	14.10	80.2

Simulation Trace (Excerpt)

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Reference scenario: 2-Purlin\nScenario summary table omitted for brevity.
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